



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF1-146

HOYTDALÉ -MCCLELLAND 345 KV

121.17 MW Capacity / 201.95 MW Energy

September 2022

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is ATSI.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 Revision Since August 2020 System Impact Study Report

The system impact study report has been revised to reflect the stability results. AF1-146 meets the power factor requirements. See section 15 for the details.

4 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Lawrence County, Pennsylvania. The installed facilities will have a total capability of 201.95 MW with 121.17 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 31, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-146
Project Name	HOYTDAL E -MCCLELLAND 345 KV
State	Pennsylvania
County	Lawrence
Transmission Owner	ATSI
MFO	201.95
MWE	201.95
MWC	121.17
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

5 Point of Interconnection

The interconnection of the project at the Point of Interconnection (POI) will be accomplished by constructing a new 345 kV three (3) breaker ring bus substation and looping the Hoytdale-McClelland 345 kV Line into the new station. The new substation will be located approximately 5.2 miles from McClelland Substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed three breaker ring bus site. The project will also require non-direct connection upgrades at McClelland and Hoytdale substations.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AF1-146 generation project to connect to the FirstEnergy (“FE”) transmission system. IC will be responsible for constructing the facilities on its side of the POI, including the attachment facilities which connect the generator to the FE transmission system’s direct connection facilities.

6 Cost Summary

The AF1-146 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 16,484,200
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 16,484,200

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

7 Transmission Owner Scope of Work

The interconnection of the project at the POI will be accomplished by constructing a new 345 kV three (3) breaker ring bus substation and looping the Hoytdale-McClelland 345 kV Line into the new station. The new substation will be located approximately 5.2 miles from McClelland Substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed three breaker ring bus site. The project will also require non-direct connection upgrades at McClelland and Hoytdale substations.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facility: Install one span of 954 ACSR towards the customer substation	\$ 263,700
Direct Connection: Build new 345 kV three-breaker ring bus substation	\$ 8,635,600
Direct Connection: Install fiber from McClelland Substation to AF1-146 Interconnection for communication transport.	\$ 691,100
Direct Connection: Install fiber from Hoytdale Substation to fiber backbone tap on Darlington fiber build for relaying communications to AF1-146	\$ 1,274,300
Non-Direct Connection: Loop the Hoytdale-McClelland 345 kV Line into new generation interconnection substation using roughly 2 spans of 954 ACSR (approximately 5.2 miles from the McClelland Substation)	\$ 3,025,400
Non-Direct Connection: Relay panel upgrade at McClelland Substation	\$ 251,800
Non-Direct Connection: Relay settings changes at Hoytdale Substation	\$ 1,715,500
Non-Direct Connection: Engineering review	\$ 626,800
Total Physical Interconnection Costs	\$ 16,484,200

8 Schedule

Based on the scope of work for the Attachment Facilities and the Direct and/or Non-Direct Connection facilities, it is expected to take a minimum of **20 months** after the signing of an Interconnection Construction Service Agreement to complete the installation. This includes the requirement for the IC to make a preliminary payment that compensates FE for the first three months of the engineering design work that is related to the construction of the interconnection substation. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined direct connection and network upgrades, and that all transmission system outages will be allowed when requested.

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

9 Transmission Owner Analysis

9.1 Power Flow Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF1-146 project did not contribute to any overloads on the FE <100 kV transmission system.

10 Interconnection Customer Requirements

10.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE’s “Requirements for Transmission Connected Facilities” document located at:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE’s “Requirements for Transmission Connected Facilities” document and will not be accepted. The GSU transformer must have a grounded wye connection on the high (utility) side and a delta connection on the low (generator) side.

10.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 345 kV circuit breaker to protect the AF1-146 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF1-146 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

10.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE transmission system.

11 Revenue Metering and SCADA Requirements

11.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

11.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter²)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)
- Wind direction (decimal degrees from true north) – (Accepted, not required)

11.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

12 Summer Peak Analysis

The Queue Project AF1-146 was evaluated as a 201.9 MW (Capacity 121.2 MW) injection tapping the Hoytdale to Shenango 345 kV line in the ATSI area. Project AF1-146 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-146 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

12.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

12.4 Steady-State Voltage Requirements

To be determined in Facilities Study

12.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

12.6 System Reinforcements

None.

12.7 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

None

13 Light Load Analysis

Not required for solar projects.

14 Short Circuit Analysis

The following Breakers are overdutied:

None

14.1 System Reinforcements - Short Circuit

None

15 Stability and Reactive Power

AF1-146 meets the power factor requirements.

Generator Interconnection Request AF1-146 is for a 201.95 MW Maximum Facility Output (MFO) solar generating facility, which consists of 64 x 3.1996 MW Power Electronic FS3430 PV inverters interconnecting with the ATSI transmission system. The AF1-146 solar generating facility will be located in Lawrence County, Pennsylvania.

The interconnection of the project at the Point of interconnection (POI) will be accomplished by constructing a new 345 kV three breaker ring bus substation and looping the Hoytdale - McClelland 345 kV line into the new station. The new substation will be located approximately 5.2 Miles from McClelland substation. The Point of Interconnection (POI) will be where the Interconnection Customer gen-tie line terminates at the new substation.

This report describes a dynamic simulation analysis of AF1-146 as part of the overall system impact study. The load flow scenario for the analysis was based on the RTEP 2023 peak load case, modified to include applicable queue projects. AF1-146 has been dispatched online at maximum power output, with approximately 1.000 pu voltage at the generator terminals.

AF1-146 was tested for compliance with NERC, PJM, Transmission Owner, and other applicable criteria. 97 contingencies were studied, each with a 20 second simulation time period (with 1.0 second initial run prior to any events). Studied faults included:

- a) Steady state operation (Category P0);
- b) Three phase faults with normal clearing time on the intact network (Category P1);
- c) Single phase to ground faults with delayed clearing due to a stuck breaker (Category P4);
- d) Single phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from the fault due to primary communications/relay failure (Category P5);
- e) Single phase to ground faults with normal clearing for common structure (Category P7).

For all 97 fault contingencies tested on the 2023 peak load case:

- a) AF1-146 was able to ride through the faults.
- b) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No other transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

Please note that the solar generating facility AF1-146 is higher than the MFO at the POI. A fictitious load of at least 1.26 MW should be added to limit the MFO at POI to the requested MFO.

16 Affected Systems

16.1 NYISO

None

16.2 MISO

None

17 Attachment 1: One Line Diagram

